

OSVVM Settings

User Guide

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1 OSVVM Settings Overview

OSVVM supports Tcl script settings in `OsvvmSettingsLocal.tcl` and VHDL settings in `OsvvmScriptSettingsPkg.vhd` and `OsvvmSettingsPkg.vhd`.

The default OSVVM settings are intended to be the best settings for new users of OSVVM. So, if you are just getting started with OSVVM, you can skip reading this document for now.

2 OSVVM Settings Directory for VHDL and Script Settings

The OSVVM Settings Directory for VHDL and Script settings is found in the directory specified by joining the Tcl variables `$SettingsRootDirectory`, `$SettingsDirectory`, and `$::osvmm::OsvvmSettingsSubDirectory`.

The `SettingsRootDirectory` is `$OsvvmLibraries`. `$OsvvmLibraries` is the location of your `OsvvmLibraries` install. `$OsvvmLibraries` is a Tcl variable set by the scripts when started with `$OsvvmLibraries/Scripts/StartUp.tcl`.

The `SettingsDirectory` is determined by the first of the following found:

- The Tcl variable `$::osvmm::OsvvmSettingsDirectory` - (highest precedence)
- Environment variable `OSVVM_SETTINGS_DIR`
- Directory `$OsvvmLibraries/./OsvvmSettings`
- Directory `$OsvvmLibraries/osvmm` for vhd settings and `$OsvvmLibraries/Scripts` for script settings - (lowest precedence)

The `OSVVM_SETTINGS_DIR` variable can contain either an absolute path or a relative path. If `OSVVM_SETTINGS_DIR` contains an absolute path, then the path of `$OsvvmLibraries` in the join is ignored.

`$::osvmm::OsvvmSettingsSubDirectory` is set by `OsvvmSettingsLocal.tcl`. As a result, this value is "" for scripts (since `OsvvmSettingsLocal.tcl` has not been called yet). If multiple simulators are being used, set `OsvvmSettingsSubDirectory` as follows:

```
set OsvvmSettingsSubDirectory ${ToolName}
```

3 OsvvmScriptSettingsPkg

OsvvmScriptSettingsPkg provides Tcl project settings to the VHDL code in the OSVVM Utility Library. It uses deferred constants that are declared in the package declaration OsvvmScriptSettingsPkg.vhd. Typically deferred constants get their value in the package body OsvvmScriptSettingsPkg_generated.vhd.

3.1 OsvvmScriptSettingsPkg_generated.vhd

The values in the generated package body are derived from the Tcl settings. Most ensure that the Tcl scripts and OSVVM Utility library are using the same file names and paths.

OSVVM_SETTINGS_REVISION is important. It is used in OsvvmSettingsPkg_default.vhd to pick between new and old (backward compatible) settings in the OSVVM Utility library. If you need to update the OSVVM libraries during a project, you can set the Tcl variable ::osvvm::OsvvmVersionCompatibility to the version of the OSVVM version when the project started to maintain compatible settings.

```
-- This file is autogenerated by CreateOsvvmScriptSettingsPkg
package body OsvvmScriptSettingsPkg is
    constant OSVVM_HOME_DIRECTORY      : string := "C:/OsvvmLibraries" ;
    constant OSVVM_RAW_OUTPUT_DIRECTORY : string := "" ;
    constant OSVVM_BASE_OUTPUT_DIRECTORY : string := "" ;
    constant OSVVM_BUILD_YAML_FILE      : string := "OsvvmRun.yml" ;
    constant OSVVM_TRANSCRIPT_YAML_FILE : string := "OSVVM_transcript.yml" ;
    constant OSVVM_REVISION              : string := "2026.08" ;
    constant OSVVM_SETTINGS_REVISION    : string := "2026.08" ;
end package body OsvvmScriptSettingsPkg ;
```

3.2 Package body options

If for some reason, these values are not appropriate, you may override them by creating a package named OsvvmScriptSettingsPkg_local.vhd in the Settings Directory. If the package OsvvmScriptSettingsPkg_generated.vhd could not be created (due to directory or file permissions), the package OsvvmScriptSettingsPkg_default.vhd (in OsvvmLibraries/osvvm) will be used.

4 OsvvmSettingsPkg

OsvvmSettingsPkg defines constants that used by the OSVVM Utility Library. Mostly these constants are used to allow changes in behavior to be made to the OSVVM utilities while supporting alternate settings to maintain backward compatibility.

These settings are important as improving the quality of the random seed hash functions will change the results produced by all tests cases that use randomization. Having these change late in a project when test cases are stable and producing required coverage would be bad. Having these change at the beginning of a new project has no impact at all.

Note many of these settings are formulated using OSVVM_SETTINGS_REVISION, a constant from OsvvmScriptSettingsPkg. OSVVM_SETTINGS_REVISION in turn is generated from the Tcl variable ::osvvm::OsvvmVersionCompatibility.

4.1 Package Body and Interface Groups

OsvvmSettingsPkg_default.vhd is a package body that sets deferred constants. It is organized as follows.

```
package body OsvvmSettingsPkg is
  -- Settings shared by all packages
  -- Settings for Requirements Tracking
  -- Settings for RandomPkg
  -- Settings for ScoreboardGenericPkg
  -- Settings for CoveragePkg
  -- Settings for AlertLogPkg
end package body OsvvmSettingsPkg ;
```

Note that there are additional constants, but changing them will require a coordinated change in OSVVM scripting, so it is not recommended.

4.2 Shared Settings for All Packages

Constant Name	Description	Value
OSVVM_HEADER_PREFIX	Default prefix for LogHeader, PrintHeader	" "
OSVVM_HEADER_SUFFIX	Default suffix for LogHeader, PrintHeader	"*"
OSVVM_LINE_LENGTH	Number of times a character repeats in a header line – LogHeader, PrintHeader, HeaderLine	162
OSVVM_LINE_WRAP	Length at which to wrap a line. To disable set to integer'high. Used for MultiLine printing in Alert, Log, PrintLine, ...	162

Constant Name	Description	Value
OSVVM_DEFAULT_TIME_UNITS	Default time units for printing	1 ns
OSVVM_DIGITS_FOR_TIME_FRACTION	Digits to print in a time fraction – see format	3
OSVVM_MAX_DIGITS_FOR_FIXED_POINT_REAL	Digits to print in fixed point – see format Switches to exponent notation after this	8
OSVVM_DIGITS_FOR_REAL_FRACTION	Fraction digits for real – see format	4
OSVVM_MAX_TIME_DECIMAL_DIGITS	Size of string used in format for time. Max number of 10**digits in 2**62 – 1 = 19 digits plus 1 for decimal point plus 2 for margin.	22

4.3 Settings for Requirements Tracking

Constant Name	Description	Value
ALERT_LOG_DEFAULT_PASSED_GOAL	Value used if requirement goal not specified when using ReadSpecification to read a requirements specification	1
COVERAGE_REQUIREMENT_BY_BIN	if TRUE, each bin of a coverage model is one requirement, otherwise, an entire coverage model is one requirement	TRUE

4.4 Settings for RandomPkg

Constant Name	Description	Value
RANDOM_USE_NEW_SEED_METHODS	Selects higher quality seed generation methods for InitSeed. Make it TRUE for new designs. 2024 revisions and later Before 2024	TRUE FALSE

4.5 Settings for ScoreboardGenericPkg

Constant Name	Description	Value
SCOREBOARD_YAML_IS_BASE_FILE_NAME	When TRUE, WriteScoreboardYaml only requires a base name and fills in the rest of the required naming pattern. 2024 revisions and later Before 2024	TRUE FALSE

4.6 Settings for CoveragePkg

Constant Name	Description	Value
COVERAGE_DEFAULT_WEIGHT_MODE	Selects weight mode to use for randomization. 2024 revisions and later Before 2024	"REMAIN" "AT_LEAST"
COVERAGE_USE_NEW_SEED_METHODS	Selects higher quality seed generation methods for InitSeed.	TRUE

Constants for WriteBin reports. Not relevant for HTML reports.

Constant Name	Description	Value
COVERAGE_WRITE_PASS_FAIL		FALSE
COVERAGE_WRITE_BIN_INFO		TRUE
COVERAGE_WRITE_COUNT		TRUE
COVERAGE_WRITE_ANY_ILLEGAL		FALSE

4.7 Settings for AlertLogPkg

4.7.1 Enable / Disable Alerts at Startup

Constant Name	Description	Value
ALERT_LOG_GLOBAL_ALERT_ENABLE	If FALSE mask all alerts. Only use at startup. Different from disabled as disabled Alerts are still counted.	TRUE

4.7.2 Controls for AlertIfDiff, AffirmIfFilesMatch, AffirmIfTranscriptsMatch

Constant Name	Description	Value
ALERT_LOG_IGNORE_SPACES	On file comparisons, ignore differences in spaces	FALSE
ALERT_LOG_IGNORE_EMPTY_LINES	On file comparisons, ignore differences in blank lines	FALSE

4.7.3 Defaults for Alert Stop Counts

Constant Name	Description	Value
ALERT_LOG_STOP_COUNT_FAILURE		1
ALERT_LOG_STOP_COUNT_ERROR	VUnit sets this to 1	Integer'high
ALERT_LOG_STOP_COUNT_WARNING		Integer'high

4.7.4 Defaults for Log Enables

Constant Name	Description	Value
LOG_ENABLE_INFO	VUnit sets this to TRUE	FALSE
LOG_ENABLE_DEBUG		FALSE
LOG_ENABLE_PASSED		FALSE
LOG_ENABLE_FINAL		FALSE

4.7.5 Control what is a test failure

Constant Name	Description	Value
ALERT_LOG_FAIL_ON_WARNING		TRUE
ALERT_LOG_FAIL_ON_DISABLED_ERRORS		TRUE
ALERT_LOG_FAIL_ON_REQUIREMENT_ERRORS		TRUE
ALERT_LOG_FAIL_ON_VHDL_ASSERT_ERRORS	With 2026 releases and newer Before 2026	TRUE FALSE
ALERT_LOG_PRINT_VHDL_ASSERT_ERRORS	With 2026 releases and newer Before 2026	TRUE FALSE

4.7.6 Alert/Log Detailed Printing Control

Constant Name	Description	Value
ALERT_LOG_WRAP	Turn on line wrap by default	FALSE
ALERT_LOG_WRAP_INDENT_CHAR	If at start of message, start message on the next line at an indent of ALERT_LOG_INDENTED_LENGTH and enable line wrapping independent of	'<'

Constant Name	Description	Value
	ALERT_LOG_WRAP setting.	
ALERT_LOG_WRAP_END_CHAR	If at start of message, print message starting after the Alert/Log prints and enable line wrapping independent of ALERT_LOG_WRAP setting	'>'
ALERT_LOG_INDENTED_LENGTH	Number of characters to indent for an indented wrap line. 21 is aligns with printing Alert/Log	21
ALERT_LOG_JUSTIFY_ENABLE	Align printing of the message at the end of the Alert/Log For releases 2024 and later Releases before 2024	TRUE FALSE
ALERT_LOG_TIME_JUSTIFY_AMOUNT	Amount of space for printing time value For releases 2026.08 and later Otherwise	16 9
ALERT_LOG_DIGITS_FOR_TIME_FRACTION	Number of time fraction digits to print with Alert/Log	1
ALERT_LOG_DIGITS_FOR_REAL_FRACTION	Number of real fraction digits to print in alert/log	4
ALERT_LOG_WRITE_ERRORCOUNT	Write the current error count immediately after OSVVM_PRINT_PREFIX AKA Mike Pagen Mode	FALSE
ALERT_LOG_WRITE_TIME_FIRST	In Alert/Log print time first For releases 2024 and later or if ALERT_LOG_WRAP is true Otherwise	TRUE FALSE
ALERT_LOG_WRITE_NAME	Print Alert / Log	TRUE
ALERT_LOG_WRITE_LEVEL	Print Alert / Log level	TRUE
ALERT_LOG_WRITE_TIME	Print time value in message	TRUE
ALERT_LOG_ID_SEPARATOR	Characters between a parent name and child name	": "
ALERT_LOG_SPACES_BEFORE_ALERT	For releases 2026.08 and later Otherwise	" " " "

Constant Name	Description	Value
ALERT_LOG_SPACES_BEFORE_LEVEL	For releases 2026.08 and later Otherwise	" " " "
ALERT_LOG_SPACES_BEFORE_ID_NAME	For releases 2026.08 and later Otherwise	"" " "

4.7.7 ReportAlerts Controls. Not relevant if using HTML reports

Constant Name	Description	Value
ALERT_LOG_REPORT_HIERARCHY	Print detailed report when fail	TRUE
ALERT_LOG_PRINT_PASSED	Print passed count in summary line	TRUE
ALERT_LOG_PRINT_AFFIRMATIONS	Print affirmations for each level	FALSE
ALERT_LOG_PRINT_DISABLED_ALERTS	Print disabled alerts for each level	FALSE
ALERT_LOG_PRINT_REQUIREMENTS	Print requirements always	FALSE
ALERT_LOG_PRINT_IF_HAVE_REQUIREMENTS	Print requirements if have any	TRUE

4.8 Overriding the Default OsvvmSettings package

The version of OsvvmSettingsPkg used is determined by the first of the following found:

- \${SettingsDirectory}/OsvvmSettingsPkg_local.vset - (highest precedence)
- \${SettingsDirectory}/OsvvmSettingsPkg_local.vhd
- Directory \$OsvvmLibraries/osvvm/OsvvmSettingsPkg_default.vhd - (lowest)

4.9 Using OsvvmSettingsPkg_local.vset

OsvvmSettingsPkg_local.vset is a file of settings. First item is the constant name, followed immediately by a ":", and then the value. The value can be a VHDL expression. To change the Alert/Log spacing settings back to the previous settings, one can do the following in their OsvvmSettingsPkg_local.vset.

```
ALERT_LOG_TIME_JUSTIFY_AMOUNT: 9
ALERT_LOG_DIGITS_FOR_TIME_FRACTION: 0
ALERT_LOG_SPACES_BEFORE_ALERT: " "
ALERT_LOG_SPACES_BEFORE_LEVEL: " "
ALERT_LOG_SPACES_BEFORE_ID_NAME: " "
```

Only the settings that need to be changed are needed in the file. The value to the right of the ": " should be a legal VHDL value that is appropriate a constant of its type.

To apply the settings, the scripts read `OsvvmSettingsPkg_default.vset` to establish the default settings. Then `${SettingsDirectory}/OsvvmSettingsPkg_local.vset` is read to override the default settings. `${SettingsDirectory}/OsvvmSettingsPkg_generated.vhd` is created by substituting the settings into `OsvvmSettingsPkg_template.vhd`. The scripts then analyze `${SettingsDirectory}/OsvvmSettingsPkg_generated.vhd`.

4.10 Using OsvvmSettingsPkg_local.vhd

In this method, `OsvvmSettingsPkg_local.vhd` is created by copying `OsvvmSettingsPkg_default.vhd` and changing the desired values. It is simple in concept, but difficult to maintain. When new settings are added to `OsvvmSettingsPkg_default.vhd` they also need to be added to the local package also.

The `OsvvmSettingsPkg_local.vset` approach only changes the items that need to change and only needs to be updated if a new OSVVM version adds a setting that needs to be changed.

5 Variables Set by Simulator Scripts

The following tcl variables are set by `VendorScripts_<ScriptBaseName>.tcl`. Where `ScriptBaseName` is a tcl variable that uniquely identifies the tool being run.

```
# variable ToolVendor
# variable ToolName
# variable ToolVersion
# variable ToolNameVersion
# variable ToolArgs
# variable NoGui
# variable ToolSupportsGenericPackages
# variable ToolType
```

6 OsvvmSettingsLocal.tcl

`OsvvmSettingsLocal.tcl` is used to configure script settings. A template is provided in the `OsvvmSettingsLocal_Example.tcl` in directory `OsvvmLibraries/Scripts`. See the next section Settings Directory for where you should put `OsvvmSettingsLocal.tcl`.

6.1 Structure of OsvvmSettingsLocal.tcl

```
# Global Settings
namespace eval ::osvvm {
# OSVVM Name Space Settings
# Version Compatibility Settings
# VHDL File Viewer Settings
```

```

# Directory structure and results file management
# TCL Error signaling during a build
# Stop Counts for Failures seen by Analyze and Simulate
# Generate transcripts
# VHDL Simulation Settings
# Default Coverage Options
# Simulation Controls
# Extended Analyze and Simulate Options
# GHDL Analyze and Simulate Options
# Second Top
}

```

6.2 Global Settings

Set variables to your project variables here.

```
variable Proj1 C:/Dev/Proj1
```

This will allow you to build your project as:

```
build $Proj1/build.pro
```

6.3 Version Compatibility Settings (OSVVM_SETTINGS_REVISION)

Version compatibility allows OSVVM's settings to work like previous versions.

```
variable OsvvmVersionCompatibility $OsvvmVersion ;# 2023.00
```

Generate failure on status NOCHECKS (when 1), otherwise, passed.

```
variable FailOnNoChecks 1 ;# 0 ;# false
```

Use the old version of CreateClock and CreateReset procedures.

```
variable ClockResetVersion $OsvvmVersion ;# 2023.00
```

6.4 VHDL File Viewer Settings

In the Test Case report, a link to the VHDL test case file is provided. Set variable VhdlFileViewerPrefix to "vscode://file/" to view the file in vscode. The default value, "", views the file in the html viewer.

```

# variable VhdlFileViewerPrefix      "" ; # Default
# variable VhdlFileViewerPrefix      "vscode://file/" ; # view with vscode

```

6.5 Directory Structure and Results File Management

Base directory to contain other outputs.

```
variable OutputBaseDirectory "" ;# "osvvm"
```

Subdirectories of the base directory.

```
variable LogSubdirectory      "logs"      ;# logs in $OutputBaseDirectory/logs
variable ReportsSubdirectory  "reports"   ;# reports in $OutputBaseDirectory/reports
variable ResultsSubdirectory  "results"   ;# results in $OutputBaseDirectory/results
variable CoverageSubdirectory "CodeCoverage" ;# Code Coverage in . . .
```

Formulation of Library Directory

```
# Library Directory = [file join $VhdlLibraryParentDirectory \
#                      $OutputBaseDirectory $VhdlLibraryDirectory $VhdlLibrarySubdirectory]
variable VhdlLibraryParentDirectory "" ;# "C:/tools"
variable VhdlLibraryDirectory      "VHDL_LIBS"
variable VhdlLibrarySubdirectory    "${ToolNameVersion}" ;# default value
```

Directory to hold temporary files. Typically same as OutputBaseDirectory.

```
variable OsvvmTemporaryOutputDirectory "" ;# temporary files
```

6.6 Supporting Different VHDL Settings for Different Simulators

Setting OsvvmSettingsSubDirectory adds a subdirectory to the search path or generated file path for VHDL settings. For details see [Settings Directory](#). A good setting for supporting multiple simulators is:

```
variable OsvvmSettingsSubDirectory $ToolName
```

If multiple simulators are not being used, the default value of "" is just fine.

6.7 Supporting Different Script Settings for Different Simulators

OsvvmSettingsSubDirectory cannot be used to select a simulator specific version of OsvvmSettingsLocal.tcl since OsvvmSettingsLocal.tcl sets OsvvmSettingsSubDirectory – a chicken and egg problem.

Instead, in addition to running OsvvmSettingsLocal.tcl (if it exists), StartUpShared.tcl will also run OsvvmSettingsLocal_<ScriptBaseName>.tcl (if it exists).

If different naming than <ScriptBaseName> is needed, then OsvvmSettingsLocal.tcl can include something like the following to run simulator specific settings:

```
include OsvvmSettingsLocal_${ToolName}.tcl
```

Small customizations can be added to OsvvmSettingsLocal.tcl using conditionals such as the following:

```
if {($ToolVendor eq "Siemens") || ($ToolVendor eq "Aldec")} {
    variable ExtendedAnalyzeOptions "-quiet"
    variable ExtendedSimulateOptions "-quiet"
}
```

6.8 What Should Generate TCL Errors

```
variable FailOnBuildErrors      "true"      ;# simulator command had errors
variable FailOnReportErrors     "false"     ;# yaml reports caused html failure
variable FailOnTestCaseErrors   "false"     ;# one or more test case(s) had errors
variable FailOnEmptyTestSuite   "true"
variable FailOnVhdlNameNotMatchTestName "true"
```

6.9 Stop Build on Number of Analyze and Simulate Errors

Stop build when AnalyzeErrorStopCount and SimulteErrorStopCount errors have occurred. If the value is 0, then do not stop. Do not stop is appropriate for regression runs.

```
variable AnalyzeErrorStopCount  0
variable SimulateErrorStopCount 0
```

6.10 Type of Transcripts to Generate

Generate HTML Transcript

```
variable TranscriptExtension     "html"      ;# Generate log and html transcripts
```

Generate Simulation Script for entire run

```
variable CreateSimScripts       "false"     ;# Create a simulator script
```

Generate Log file with just OSVVM Output

```
variable CreateOsvvmOutput      "false"     ;# log file with just OSVVM output
```

6.11 Select MemoryPkg

UseMemoryPkg01 TRUE results in selecting 01 policy for MemoryPkg, otherwise, selects UX01 policy. Policy 01 stores only values 0 and 1 for if the memory is > 16 bits, cuts the memory storage in half. Policy UX01 stores values U, X, 0, and 1 – resulting in a more accurate simulation.

```
variable UseMemoryPkg01        "false"
```

6.12 Requirement Tracking Settings

Setting USE_SUM_OF_GOALS when false, uses maximum goal - good when merging in specification which provides the maximum goal which is divided across tests, otherwise, when true, uses sum of goals - good when not merging the specification and need to sum up goals to get the total.

```
variable USE_SUM_OF_GOALS      "false" ;# when false, uses maximum
```

Setting REQUIREMENT_CSV_PRINT_STATUS when false, do not print status in CSV, otherwise, when true, print status in CSV

```
variable REQUIREMENT_CSV_PRINT_STATUS "false" ;# do not print Status in CSV
```

Setting REQUIREMENT_TEST_CASE_FAILS_IF_LESS_THAN_GOAL when false, test case status remains unchanged (Errors=0 => PASSED), otherwise, when true, if requirements < goal, change test case status to FAILED

```
variable REQUIREMENT_TEST_CASE_FAILS_IF_LESS_THAN_GOAL "true"
```

Setting REQUIREMENT_DOES_NOT_EXCEED_GOAL when false, sum up actual requirement value, otherwise, when true, requirement = Minimum(Requirement, Goal)

```
variable REQUIREMENT_DOES_NOT_EXCEED_GOAL "true" ;# do not print Status in CSV
```

6.13 VHDL Simulation Settings

```
# Valid VHDL Version values 1993, 2002, 2008, 2019
variable DefaultVHDLVersion "2008" ; # OSVVM requires >= 2008.
variable SimulateTimeUnits "ps"
variable DefaultLibraryName "DefaultLib"
```

6.14 Default Coverage Options

```
variable CoverageEnable "true"
variable CoverageAnalyzeEnable "false"
variable CoverageSimulateEnable "false"
variable CoverageAnalyzeOptions [vendor_SetCoverageAnalyzeDefaults]
variable CoverageSimulateOptions [vendor_SetCoverageSimulateDefaults]
```

6.15 Default Simulation Controls

```
variable SimulateInteractive "false"
variable DebugIsSet "false"
variable Debug "false"
variable LogSignalsIsSet "false"
variable LogSignals "false"
variable ScriptDebug "false"
```

6.16 Extended Analyze and Simulate Options

```
variable VhdlAnalyzeOptions ""
variable VerilogAnalyzeOptions ""
variable ExtendedAnalyzeOptions "" ;# "-quiet"
variable ExtendedSimulateOptions "" ;# "-quiet"
```

6.17 GHDL and NVC Analyze and Simulate Options

```
variable ExtendedElaborateOptions ""
variable ExtendedRunOptions ""
variable SaveWaves "false"
variable SimulateInteractive "false"
```


6.18 OSVVM Developer Controls

```
variable TclDebug "false" ;# print $::errorInfo on failures
variable ReportDebug "false" ;# also print on report failures
variable OsvvmDevDeriveArchitectures "false" ;# Derive 2008 arch from 2019
variable OsvvmDevDerivePackageHeaders "false" ;# Derive OsvvmSettingsPkg
```

6.19 Name of Second Top Level

```
variable SecondSimulationTopLevel ""
```

6.20 Deprecated Settings

The setting `SettingsAreRelativeToSimulationDirectory` is deprecated and the only valid value is "false". The value will be "false" in `StartUpShared`, since `OsvvmSettingsLocal.tcl` has not run yet. As a result, if set to "true", VHDL settings will be in a different directory from the script settings. In a future revision of OSVVM, it is likely that setting this to "true" will result in a tcl error.

7 About the Author - Jim Lewis

Jim Lewis, the founder of SynthWorks, has thirty plus years of design, teaching, and problem solving experience. In addition to working as a Principal Trainer for SynthWorks, Mr Lewis has done ASIC and FPGA design, custom model development, and consulting.

Mr. Lewis is the primary developer of the Open Source VHDL Verification Methodology (OSVVM.org) packages and a participant in the IEEE VHDL standards. Neither of these activities generate revenue.

Please support our volunteer efforts by buying your VHDL training from SynthWorks.

If you find bugs these packages or would like to request enhancements, you can reach me at jim@synthworks.com.